Structural Self-Energy Expansion (SSEE)

A minimal-parameter dark energy framework from φ and π

Mike Edison Almeida Vallejo ORCID: 0009-0008-2195-7836
mike.almeida1721@gmail.com github.com/mikealmeida1721/SSEE-Cosmologia

Seeking arXiv endorsement in astro-ph.CO (or gr-qc)

Core idea in one sentence

The core cosmological background — w_0, w_a, H_0, Ω_m, n_s — is derived *algebraically* from the golden ratio $\varphi = (1+\sqrt{5})/2$ and π with **zero fitted dimensionless parameters at the background level**. The kineticity $\alpha_K = 15.591336$ of the dark-energy condensate (Paper 7) is derived, not fitted, and is unobservable on linear scales. The dark-sector ansätze that previously appeared here (m_φ, ξ) were retracted in 2026-08 together with the second matter sector, and their register entries (OP-9 through OP-12) close by dissolution; the MIRA matter mechanism (OP-8) had already dissolved under the ω_m-direct reframe. The remaining open entries are listed in `OPEN_PROBLEMS.md`.

Key algebraic predictions

Quantity	SSEE algebraic value	Observed / constraint	Agr
w_0	$-T_r/M_v = -0.8400$	DESI DR2: -0.838 ± 0.055	0.24
w_a	$-P_{sc}/I_g = -0.669975$	DESI DR2: -0.631 ± 0.226	0.17
$3(\varphi+\pi)^2$ (target)	pure number $= 67.96214$	Planck 2018: 67.36 ± 0.54	1.11
$\Omega_{m,\mathrm{CMB}}$	$\omega_m/h^2 = 0.308881$ (ω_m-direct)	Planck 2018: 0.3153 ± 0.0073	0.88
r_d	CAMB (ω_m-direct): 147.17 Mpc	Planck 2018: 147.09 ± 0.26 Mpc	0.32
H_0^{glob}	$H_0^{\mathrm{SH0ES}}(1-f_{\mathrm{scr}}) = 67.962142$ km/s/Mpc	$3(\varphi+\pi)^2 = 67.96214$	resi
n_s	$1-\varphi^{-7} = 0.965558$	Planck 2018: 0.9649 ± 0.0042	0.16
r	$\varphi^{-10} = 0.008131$	BK18: $r < 0.036$	Pre
c_s^2	$(M_v - T_r)/(5M_v + 3T_r) = 0.021284$	dark-energy clustering (Euclid)	Pre
S_8	0.8273 (predicted; *no* free cosmological parameter)	KiDS-Legacy: 0.8265 ± 0.0176	0.04

Ten-paper series (all reproducible)

I. Derived Algebraic Background

- **Paper 1 (Framework)**: Algebraic derivation of geometric constants; α-attractor embedding ($\alpha = \varphi^4/3$). [35 pp]
- **Paper 4 (Nine Sovereignties)**: CMB observables (ℓ_1, ℓ_2, ℓ_3, θ_*, A_s, n_s, τ, r) bounded algebraically. [19 pp]

II. Tested Against Data

- **Paper 2 (MCMC)**: 100 walkers × 25 000 steps vs. DESI DR2 + Planck (the cluster term was removed: cluster masses are tested separately, in general relativity, against the 46 systems of Zhang et al. 2026); $\Delta\mathrm{BIC} = -7.25$ (SSEE favoured, $k_{\mathrm{SSEE}} = 2$ vs. $k_{\Lambda\mathrm{CDM}} = 3$), total matter density in the background geometry. [31 pp] Convention throughout this summary: $\Delta \equiv \mathrm{SSEE} - \Lambda\mathrm{CDM}$; negative Δ favours SSEE.
- **Paper 3 (CMB Confrontation)**: CAMB ω_m-direct ($\Omega_m = 0.308881$); Cobaya full `plik` MCMC; $\Delta\mathrm{BIC} = -33.83$ (canonical; -26.03 in the `plik_lite` point estimate with $\{A_s, \tau\}$ fitted and $N = 669$ measured from the likelihood), SSEE favoured under $k = 2$ vs $k = 6$ model comparison. [26 pp]

- **Paper 6 (Growth sector, against raw $\xi_\pm$)**: retracts an earlier two-sector φ-DM proposal and its particle — the subtraction defining the second sector mixed a measured density with an equation-of-state number, and the S_8 tension motivating it exists only in the compressed statistic, whose reduction assumes ΛCDM. Against the **KiDS-Legacy** release (357 raw points; Wright et al. 2025, arXiv:2503.19441 — *published sixteen months before this analysis: no temporal priority is claimed*) the shear asks for $\log(10^{10}A_s) = 3.0255 \pm 0.0396$ against the 3.04483 that the CMB fixes under the *same* algebraic background: 0.49σ (Planck background, held equally rigid: 1.01σ). Fixing A_s to that CMB value therefore costs nothing — $\chi^2 = 417.97$ with **eight free parameters, all nuisance, none cosmological**, against 418.34 with A_s free (ΔBIC = +6.24) and 418.95 for the ΛCDM-background control (+6.86). $S_8 = 0.8273$ is then predicted, not fitted, against 0.8265 ± 0.0176 measured: 0.04σ. The three χ^2 span 1.0, so the shear does *not* discriminate between the models; what separates them is the parameter count. (Earlier, against KiDS-1000: $S_8 = 0.7559 \pm 0.0189$ with A_s free, matching the ΛCDM control 0.7571 ± 0.0194 run through the same pipeline with four more free parameters.) [17 pp]

III. Phenomenological Extensions & Open Directions

- **Paper 5 (Israel–Stewart Perturbations)**: Causal bulk viscosity; $c_s^2 = 0$ as the minimal-viscosity solution (OP-21); $\gamma_{\rm IS} = 0.5504 \approx \gamma_{\Lambda\rm CDM}$. [31 pp]
- **Paper 7 (Canonical EFT Proposal)**: the action is a minimally coupled ghost condensate, $K(X) = c_1 X + c_2 X^2$, with no potential and no dark coupling; $\alpha_T = \alpha_M = \alpha_B = 0$, $\alpha_K = 15.591336$ (not observationally accessible) and sound speed $c_s^2 = 0.021284$. The same action fixes w_0 but not w_a (OP-24). [17 pp]

IV. Strong-Field Regime, Hubble Tension & UV Completion

- **Paper 8 (Strong Gravity)** [*draft; its canonical lensing prediction rests on $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ and $\alpha_B = \alpha_M = 0$ (Paper 7), and is unconditional*]: Disformal null geodesic analysed in two limits: alternative MOND-like (deflection $\sqrt{\mathrm{AURA}} \approx 1.9995$, numerically close to $\mathcal{M} = \mathrm{AURA}/2$ at 0.03% but algebraically distinct) and the canonical limit with EFT $\alpha_B = \alpha_M = \alpha_T = 0$ (residual fifth-force $\sim 10^{-15}$ at kpc, observable lensing recovers GR); selective coupling satisfies solar-system tests. Current SLACS/BELLS data favour the canonical limit (OPEN_PROBLEMS.md OP-13). A dimensional defect in the Vainshtein-radius equation of this paper is open and under repair (OP-4, reopened 2026-09); it does not enter the canonical-limit prediction, which rests on ω_c and on $\alpha_B = \alpha_M = 0$. [19 pp]
- **Paper 9 (Hubble Tension)**: Algebraic local screening fraction $f_{\rm scr}^{\rm IR} = (\pi - \varphi)/(\varphi + \pi)^2 = 0.067253$ applied to the measured SH0ES rate 73.04 ± 1.04 returns, with the IR term alone, $H_0^{\rm glob,IR} = 68.13$ km/s/Mpc, 0.17σ from the pure algebraic number $3(\varphi + \pi)^2 = 67.96214$, the comparison target (SH0ES is the input; no fitted parameters) (2026 reframe; the earlier CMB-mapped $H_0^{\rm MIRA} = 67.04$ is superseded). [20 pp]
- **Paper 10 (UV Completion)**: Higher-derivative operator $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 9.68$ meV; connects the dark-energy EFT to the inflationary α-attractor. [16 pp]

Falsifiability (explicit)

- $S_8 = 0.8273$ — with the background fixed by the algebra *and* A_s fixed to the value the CMB gives under that same background, the growth sector has no free cosmological parameter left: cosmic shear placing S_8 more than 3σ away falsifies it outright, with nowhere to absorb the discrepancy (KiDS, DES, Euclid).
- $\log(10^{10}A_s)$ — the amplitude the shear asks for must keep agreeing with the one the CMB fixes. KiDS-1000 asked for 2.864 ± 0.050 (3.6σ away) and KiDS-Legacy for 3.0255 ± 0.0396 (0.49σ); the

two releases differ from each other by 2.5σ. A future release returning to the lower value reopens the sector.

- $\alpha_T = \alpha_M = \alpha_B = 0$ (Paper 7) — gravity is unmodified, $G_{\rm eff} = G_N$ and no gravitational slip; a detected slip or $\mu \neq 1$ in Euclid/LSST lensing×clustering would falsify the sector.
- $r = \varphi^{-10} = 0.008131$ — LiteBIRD ($\sim$2032) will confirm or refute the tensor-to-scalar ratio.
- A $> 3\sigma$ deviation of w_0 from -0.840 would falsify the corresponding background sector of the framework.

What I am asking for

An arXiv endorsement in `astro-ph.CO` (or `gr-qc`) to submit Papers 1–3 as the first tranche. All code, data, and LaTeX sources are public at github.com/mikealmeida1721/SSEE-Cosmologia under Apache-2.0.

I am happy to answer any technical questions or share the PDFs directly. Thank you for considering this request.

Code: Python (emcee, CAMB, CLASS), all reproducible in <30 min. **Data:** DESI DR2 (public), Planck 2018 (public), KiDS-1000 and KiDS-Legacy (public).